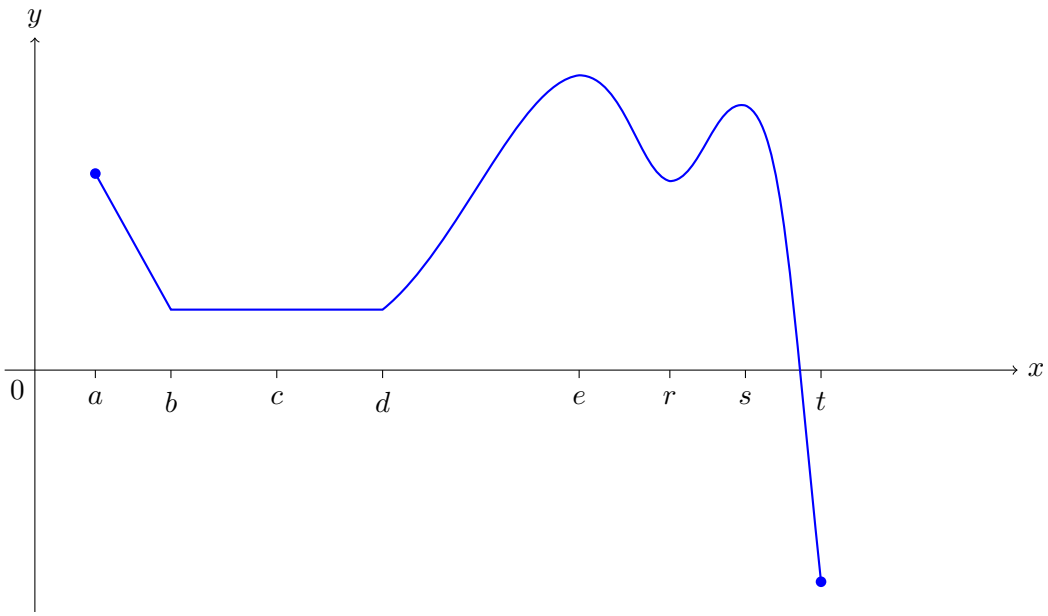
ENGSCI 111

Mathematical Modeling 1 Section 1

KP. Wang **Assignment #5** **due 8:15 am on Nov 5 (in class)**

1. For each of the numbers a, b, c, d, e, r, s, and t, state whether the function whose graph is shown has an absolute maximum or minimum, a local maximum or minimum, or neither a maximum nor a minimum.



2. Find the absolute maximum and absolute minimum values of f on the given interval.
   (a) $f(t) = t\sqrt{4 - t^2}, \quad [-1, 2]$
   (b) $f(x) = \dfrac{x}{x^2 + 4}, \quad [0, 3]$
   (c) $f(x) = 2\cos t + \sin 2t, \quad [0, \pi/2]$

3. If a and b are positive numbers, find the maximum value of
$$f(x) = x^a(1 - x)^b, \quad 0 \leqslant x \leqslant 1.$$

4. Prove that the function $f(x) = x^{101} + x^{51} + x + 1$ has neither a local maximum nor a local minimum.

5. For the following functions:
   (a) $f(x) = \sin x + \cos x, \quad 0 \leqslant x \leqslant 2\pi$
   (b) $G(x) = 5x^{2/3} - 2x^{5/3}$
      1. Find the intervals on which f is increasing or decreasing.
      2. Find the local maximum and minimum values of f.

3. Find the intervals of concavity and the inflection points.

6. Show that the function $g(x) = x|x|$ has an inflection point at $(0, 0)$ but $g''(0)$ does not exist.

7. Suppose f'' is continuous and $f'(c) = f''(c) = 0$, but $f'''(c) > 0$. Does f have a local maximum or minimum at c? Does f have a point of inflection at c?

8. Follow the given steps in class and lecture notes. Present all required information about f, f', and f''. Then sketch the given curves.
   (a) $y = x\sqrt{5 - x}$
   (b) $y = x \tan x, \quad -\pi/2 < x < \pi/2$

9. Find dy/dx by implicit differentiation.
   (a) $\tan(x/y) = x + y$
   (b) $\sqrt{x + y} = 1 + x^2y^2$

10. Find y'' by implicit differentiation.
    (a) $9x^2 + y^2 = 9$
    (b) $\sqrt{x} + \sqrt{y} = 1$

11. Find an equation of tangent line to the following curve $x^2 + y^2 = \left(2x^2 + 2y^2 - x\right)^2$ at $\left(0, \frac{1}{2}\right)$.

12. The Bessel function of order $0, y = J(x)$, satisfies the differential equation $xy'' + y' + xy = 0$ for all values of x and its value at 0 is $J(0) = 1$.
    (a) Find $J'(0)$.
    (b) Use implicit differentiation to find $J''(0)$.

13. Differentiate the following functions.
    (a) $y = (\sin x)^{\ln x}$
    (b) $y = (\ln x)^{\cos x}$

14. Some authors define $y = \sec^{-1} x \iff \sec y = x$ and $y \in [0, \pi/2) \cup (\pi/2, \pi]$. Show that with this definition, we have

$$\frac{d}{dx}\left(\sec^{-1} x\right) = \frac{1}{|x|\sqrt{x^2 - 1}}, \quad |x| > 1.$$